

## Integration — Lesson 5 Test: The Power Rule for Antiderivatives

10 questions · show your work in the box · fractions like  $x^3/3$  are welcome — keep them, don't turn them into decimals · check every answer by differentiating

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_ / 10

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1. Find the antiderivative of  $x^3$ .

1 pt

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2. Find the antiderivative of 4 (just the constant 4).

1 pt

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3. Find the antiderivative of  $9x^2$ . (Simplify your coefficient.)

1 pt

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4. Find the antiderivative of  $x^4 + x^2$ .

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5. Find the antiderivative of  $6x^5 + 2x$ .

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6. Find the antiderivative of  $x + 7$ .

1 pt

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7. Check whether  $x^6/6 + x$  is an antiderivative of  $x^5 + 1$ . Show the differentiation that proves your answer. <sup>1 pt</sup>

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8. A student says the antiderivative of  $x^2$  is  $x^3 + C$ . Use the differentiation check to show whether that's right, and fix it if needed.

1 pt

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**9.** Find the antiderivative of  $(x + 1)(x + 4)$ . Hint: multiply it out first.

1 pt

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**10.** Find  $f(x)$  given  $f'(x) = 6x^2 - 2$  and  $f(1) = 5$ . Then verify both given facts about your answer.

1 pt

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